

Mohamed Mamdouh

Full-Stack .NET & Angular Developer

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SUMMARY

Full-Stack .NET & Angular Developer with hands-on experience building scalable web applications using ASP.NET Core, Angular, SQL. Strong in Clean Architecture, secure authentication, API design, testing, and modern software development practices.

EDUCATION

Bachelor of Engineering in Computer and Systems Engineering,
Zagazig University — 2026.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, SQL, T-SQL, HTML, CSS.

Backend: ASP.NET Core Web API, Entity Framework Core, LINQ, ADO.NET, ASP.NET Identity, JWT, Refresh Tokens, HTTP-only Cookies, Role-Based Authorization (RBAC), Dependency Injection, FluentValidation, MediatR (CQRS), SignalR, Background Services, Caching, Middleware, Logging.

Frontend: Angular, RxJS, Signals, Lazy Loading, Routing, Reactive Forms, HTTP Interceptors, Route Guards, Dependency Injection, Tailwind CSS, Bootstrap.

Databases & Storage: SQL Server, PostgreSQL, MongoDB, Cloudflare R2, Database Design, Query Optimization.

Testing & DevOps: xUnit, Testcontainers, Docker, GitHub Actions (CI/CD), Postman, Swagger, Git, GitHub.

Architecture & Concepts: SOLID Principles, Design Patterns, Clean Code, Clean Architecture, 3-Tier Architecture, Domain-Driven Design fundamentals, Transactional Outbox Pattern, RESTful API Design, SDLC.

SELECTED PROJECTS

Amanah – Lost & Found Platform

[GitHub](#) [Live Production](#)

Full-stack lost-and-found platform that connects people who lost items with finders through moderated listings, structured ownership claims, and in-app chat.

Tech Stack: .NET, Angular, PostgreSQL, EF Core, FluentValidation, JWT Bearer, SignalR, Cloudflare R2, Docker, xUnit, Testcontainers, GitHub Actions.

Key Highlights:

- Implemented API reliability with centralized exception handling, correlation IDs, validation, rate limiting, HybridCache, structured logging, and dependency health checks.
- Implemented phone/email OTP registration, password authentication, short-lived JWT access tokens, and HTTP-only refresh cookies for session persistence.
- Built responsive Arabic RTL interfaces with reactive forms, EXIF metadata stripping, and client-side WebP image processing before storage uploads.
- Integrated SignalR chat with REST fallback and a transactional outbox for reliable SMS and email dispatch.
- Implemented background workers for listing expiration, claim timeouts, storage cleanup, and scheduled maintenance.
- Built integration tests with xUnit and Testcontainers using isolated database instances, and integrated them into GitHub Actions CI pipelines.

Shora – Consulting & Booking Platform

[GitHub](#) [Live Production](#)

Full-stack booking platform for one-to-one consulting sessions with availability management, manual payment verification, client bookings, and admin operations.

Tech Stack: .NET, Angular, PostgreSQL, EF Core, ASP.NET Identity, JWT Bearer, FluentValidation, Cloudflare R2, Brevo, Docker, Render, Supabase, xUnit, Testcontainers, GitHub Actions.

Key Highlights:

- Prevented double-booking with PostgreSQL row locks, transactions, and partial unique indexes, and modeled booking/payment lifecycles with state machines and audit trails.
- Built manual payment verification for Vodafone Cash/InstaPay with receipt validation, SHA-256 duplicate detection, and secure presigned URLs.
- Designed a scheduling system with recurring availability, blocked dates, booking-hold limits, payment deadlines, and cancellation rules.
- Built a role-based Angular standalone SPA using signals, lazy routes, guards, typed API contracts, and RTL, deployed with Docker, Supabase PostgreSQL, and Cloudflare R2.
- Added job monitoring, admin alerts, API error codes, and earnings reporting for gross, refunded, and net revenue.

MechanicShop – Workshop Operations API

[GitHub](#)

Backend REST API for a mechanic shop covering work orders, day scheduling, labor assignment, customer and vehicle records, and invoicing.

Tech Stack: ASP.NET Core, SQL Server, EF Core, MediatR, JWT, ASP.NET Identity, FluentValidation, Docker, xUnit, Testcontainers, GitHub Actions.

Key Highlights:

- Built a Clean Architecture backend with CQRS/MediatR, Result pattern, FluentValidation, and pipeline behaviors for validation, logging, transactions, and caching.
- Implemented secure JWT authentication with rotating refresh tokens, per-device sessions, reuse detection, RBAC, and custom policies.
- Modeled work orders, day scheduling, labor assignment, and invoicing with optimistic concurrency, with unit/API integration tests with Testcontainers and separate GitHub Actions CI jobs.

DVLD – Driver & Vehicle License Department System

[GitHub](#)

Desktop application for managing driver licenses, license applications, driving tests, and vehicle-related records.

Tech Stack: C#, .NET Framework, Windows Forms, ADO.NET, SQL Server.

Key Highlights:

- Architected a 3-tier desktop solution using C# and WinForms, enforcing clean separation between UI, domain logic, and data access layers.
- Connected to SQL Server via ADO.NET and parameterized stored procedures to execute secure entity operations and rule-driven validation queries.
- Implemented core licensing workflows, including a 3-stage exam system (scheduling, grading, retakes).